(a)

Bacteria

PCoA2 (8.5%)

PERMANOVA
A ***
F ***
C **
A × F ***
A × C *
F × C **
A × F × C *

PCoA1 (21.3%)

(b)

Fungi

PCoA2 (7.7%)

PERMANOVA
A ***
F **
C *
A × F *

PCoA1 (19.4%)

Water amount Dry Wet Watering frequency Low High Plant composition Conspecific Heterospecific

(c)

Low PCoA1 (Bacteria)
6 ASVs
5 ASVs
25 ASVs

Phylum
Actinobacteriota
Bacteroidota
Chloroflexi
Desulfobacterota
Firmicutes
Myxococcota
Proteobacteria
Spirochaetota
Verrucomicrobiota

High PCoA1 (Bacteria)
3 ASVs
2 ASVs
6 ASVs
4 ASVs
2 ASVs
9 ASVs

Representative genera
Streptomyces, Nocardioides

Representative genera
BIrii41, Turneriella

(d)

Low PCoA1 (Fungi)
6 ASVs

Phylum
Ascomycota

High PCoA1 (Fungi)
5 ASVs

Representative genera
Darksidea, Dactylella

Representative genus
Pseudomassariosphaeria